Uranium in the Desert: The Public Story of Namibia's Mining Industry

From Contested Colonial Resource to Major Global Supplier

Namibia ranks among the world's leading uranium producers, typically placing second or third after Kazakhstan and Canada and accounting for roughly 10–12 percent of global mine output in recent years. Yet the country operates no nuclear power plants, enrichment facilities, or weapons program. Its nuclear-related industry consists almost entirely of uranium mining and the export of yellowcake concentrate. Management of residual radioactivity centers on mine tailings and naturally occurring radioactive material (NORM). The Namibian Constitution explicitly prohibits the dumping or recycling of foreign nuclear and toxic waste on its territory.

Discovery and the Colonial Era

Uranium mineralization was first recognized in 1928 near the Rössing Mountains in the Namib Desert, then South West Africa under South African administration. Intensive exploration began in the late 1950s and 1960s. Rio Tinto secured rights to the large, low-grade Rössing deposit and opened the mine in 1976. It rapidly became one of the world's largest open-pit uranium operations, supplying nuclear power programs in Europe, Japan, and elsewhere.

Because the territory was under South African occupation widely regarded as illegal by the United Nations and the International Court of Justice, the mine was highly controversial. The UN Council for Namibia and anti-apartheid organizations condemned what they termed the "plunder" of Namibian uranium. Western governments and companies faced campaigns to cancel contracts. Operations continued under racially segregated conditions consistent with apartheid-era rules. Iranian interests acquired a minority stake in Rössing in the mid-1970s, a holding that has persisted. Other deposits, including Langer Heinrich and Trekkopje, were identified but remained undeveloped for decades due to price fluctuations and political conditions.

Independence and Expansion

Namibia gained independence in 1990 under the SWAPO government. The new administration treated uranium as a strategic economic asset rather than a symbol of colonial exploitation. Rössing continued operating and was rebranded as beneficial to the independent state. Production and exploration expanded significantly after the uranium price boom of the early 2000s.

The principal operating or recently restarted mines are concentrated in the Erongo Region of the central Namib:

• **Rössing** (opened 1976): The longest-running mine; majority-owned by China National Uranium Corporation (CNUC) after Rio Tinto sold its controlling stake in 2019. It retains an Iranian minority holding of approximately 15 percent and a South African Industrial Development Corporation minority stake (recently under review for exit over sanctions exposure). The Namibian government holds a minority interest.

• **Langer Heinrich** (started 2006, majority-owned by Paladin Energy): A calcrete deposit that was placed on care and maintenance during periods of low prices and restarted commercial production around 2024.

• **Husab** (production from approximately 2016): One of the world's largest uranium mines, majority-owned by China General Nuclear (CGN) via Swakop Uranium. Its discovery ranked among the most significant global uranium finds in decades.

Chinese investment has become dominant across the sector. The Namibian government holds small stakes in certain mines through state entities and has designated uranium a strategic mineral. Additional projects such as Etango, Tumas, and Norasa are in various stages of development. Uranium ranks as a top export earner for the country.

Waste, Tailings, and Disposal

Uranium mining generates large volumes of tailings that retain residual radioactivity and other contaminants. These are managed on-site under Namibian radiation protection and environmental regulations, principally the Atomic Energy and Radiation Protection Act and associated waste disposal rules. Operators are required to characterize, contain, and monitor the material, with progressive rehabilitation expected.

There is no large-scale commercial industry accepting foreign spent nuclear fuel or high-level waste. Namibia's 1990 Constitution (Article 95) directs the state to provide measures against the dumping or recycling of foreign nuclear and toxic waste on Namibian territory. Successive governments have publicly reaffirmed this position. Local radioactive sources used in medicine or industry, together with mine-related NORM, are regulated domestically. Occasional proposals for limited storage of sealed sources (for example, equipment used to calibrate oil and gas instruments) have drawn public scrutiny, but these are not equivalent to the import of foreign reactor waste.

A modern, compliant hazardous-waste facility known as Namwaste, located near Arandis, has been developed to serve the mining and industrial sectors. Gaps remain in specialized national infrastructure for low-level radioactive waste, particularly as oil and gas exploration raises additional NORM issues. Environmental and community concerns periodically surface around water use in the desert environment, dust, and the long-term stability of tailings.

The Contemporary Public Narrative

The dominant public story is one of a major, strategically important mining industry that survived the transition from contested colonial administration to independent Namibia and has since become a significant economic contributor under growing Chinese ownership and investment. The sector supplies global nuclear power markets, with a substantial share of recent output directed to China. Environmental management, water use, labor relations, and ambitions for value-addition—such as possible future local fuel fabrication—remain active topics of discussion. The constitutional ban on foreign nuclear waste dumping stands as a firm and publicly restated policy.

In summary, Namibia's nuclear story centers on large-scale, export-oriented uranium mining that began under contested colonial rule, continued and expanded after independence, and is now heavily linked to Chinese investment—accompanied by domestic regulation of mining wastes and an explicit prohibition on serving as a destination for foreign nuclear waste.

References

1. Namibian Uranium Association. “Uranium Mining in Namibia.” https://www.namibianuranium.org/uranium-mining-in-namibia/
2. World Nuclear Association. “Uranium in Namibia.” https://world-nuclear.org/information-library/country-profiles/countries-g-n/namibia.aspx
3. Wikipedia. “Uranium mining in Namibia.”
4. Ndahangwapo, Saima Nakuti. Defending the Investment: Rössing Uranium and the Business of Decolonisation in Namibia. Basler Afrika Bibliographien, 2024.
5. Hecht, Gabrielle. “The Power of Nuclear Things.” (related historical analysis of Rössing).
6. Namibian Constitution, Article 95 (environmental provisions, including prohibition on foreign nuclear and toxic waste).
7. Atomic Energy and Radiation Protection Act (Namibia) and associated Radiation Protection and Waste Disposal Regulations.
8. Recent production data from World Nuclear Association, Chamber of Mines of Namibia, and company reports (Rössing, Paladin Energy, Swakop Uranium/Husab), 2024–2026.
9. Oxpeckers Center for Investigative Environmental Journalism. Reporting on hazardous and radioactive waste infrastructure gaps in Namibia (2026).

This narrative is based on publicly available historical records, industry reports, constitutional provisions, and contemporary news coverage. It aims to present the established public story of Namibia’s uranium sector without speculation beyond documented facts.